

Paper 1.25 - Toolkit for the LCR Carrier

CQE-paper-01.25

CQECMPLX Formal Paper Series

Review-facing PDF built 2026-06-12

Construction Basis

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- top-level review source: Papers\Source\CQE-paper-01.25.md

Paper 1.25 - Toolkit for the LCR Carrier

Purpose

Paper 1.25 describes the tools available for reviewing Paper 1. The tools are support systems. They do not replace the Paper 1 proof.

Review Objects

The Paper 1 toolkit works with four objects:

```
LCR state      = (L,C,R)
center map     = pi_C(L,C,R) = C
reversal map   = rho(L,C,R) = (R,C,L)
shell grade    = L + C + R
```

The reader can verify every Paper 1 result by hand, by script, or by receipt.

Digital Toolkit

Primary executable files:

```
production/formal-papers/CQE-paper-01/verify_lcr_carrier.py
production/formal-papers/CQE-paper-01/lcr_carrier_receipt.json
```

The verifier enumerates $\{0,1\}^3$, checks reversal, counts shell grades, and rejects the false claim that all opposed boundary positions must contain unequal values.

Analog Toolkit

A physical reconstruction requires only:

- three marked positions,
- two token colors or binary marks,
- a way to flip left and right,
- a receipt line.

Procedure:

```
draw L | C | R
place binary values
record shell = L + C + R
swap L and R
check that C did not move
mark fixed state or reversal pair
test (1,0,1) against any value-inequality claim
```

The analog route exists to expose the same finite proof without requiring the software stack.

Hidden-Guess Diagnostics

When training mode is enabled, diagnostic review should ask for a prediction before revealing the receipt:

```
How many binary LCR states are there?
Does rho preserve C?
How many states are fixed by rho?
Which shell-2 state falsifies boundary-value inequality?
```

Expected answers:

8
yes
4
(1, 0, 1)

Boundary

Paper 1.25 is a toolkit supplement. It does not add claims to Paper 1. Any new claim found during tool use must be promoted through Paper 1.50's claim contract before it can enter the scientific sequence.